
A Subnet for Low-Bit Language Models

Competitive conversion and training at 1–1.58 bits

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1. Summary

This subnet produces the best low-bit version of every language model that matters, continuously and verifiably. It does so through two independent incentive mechanisms running under one subnet. In the first, miners train 1-bit and ternary models from a base model supplied by the team, and the best trainer wins. In the second, miners submit conversion code — a container that turns a full-precision model into a low-bit one without any training — and the code that produces the best model wins.

The split is not arbitrary. It mirrors how state-of-the-art low-bit models are actually built: a gradient-free stage that maps full-precision weights onto a discrete grid, followed by a training stage that recovers what the mapping destroyed. The subnet turns each stage into its own market, and lets the output of one feed the input of the other. Over time the network converges on a public, continuously improving pipeline that no single lab maintains.

2. Why low-bit models

A 16-bit 8B model occupies roughly 16 GB. The same model at 1.125 bits per weight occupies about 1.16 GB, and at 1.58 bits about 1.75 GB. That difference decides whether a model runs on a phone, a robot, a laptop, or not at all outside a datacenter. Because single-stream decoding is memory-bound, the compression also buys speed and energy: measured gains run from roughly 1.4× to 6× on CPU with 55–82% less energy, and Intel’s ultra-low-bit kernels push further on client GPUs.

The quality gap has narrowed enough to matter. PrismML’s Bonsai-8B retains about 89% of its Qwen3-8B parent’s average benchmark score at 1.125 bits; the ternary build of their 27B model retains 94.6% of its parent. But the gap is not closed, and it is unevenly distributed. At 1 bit, Bonsai 27B holds roughly 66% of parent performance on agentic tasks and 60% on vision, because per-layer rounding noise compounds across multi-step work while leaving single-shot answers largely intact. Closing that gap is the central open problem in the field.

The commercial signal is equally clear. PrismML raised a \$16.3M seed from Khosla and Cerberus, released every model under Apache 2.0, published no training code, and by July 2026 was confirmed to be in evaluation talks with Apple. The value in low-bit conversion is real, and it is currently captured through bilateral licensing between one lab and one device maker at a time.

3. Why a network is the right structure

Low-bit quality is not gated by a secret algorithm. The building blocks are public: BitNet’s BitLinear formulation and its three-stage distillation pipeline, TernaryLM’s adaptive layer-wise scaling, PT-BitNet’s post-training approach, Falcon-Edge’s fine-tunable 1.58-bit models. What separates a 90% retention model from an 85% one is the settings — which layers stay in higher precision, group size and scale estimator, per-layer thresholds and the sparsity they induce, warm-up token budget, the distillation data mix, the choice of teacher. That is a large, weakly-structured search space with an objective score at the end of it.

Parallel search against an objective score is precisely what a network does well and what a small team does poorly: a lab explores the settings one researcher’s intuition suggests, while a network explores every direction its miners can think of at once, and the score decides. No subnet currently covers low-bit work at all; Distil (SN97) explicitly rejects quantized submissions, leaving the lane open.

4. Architecture

The subnet runs two incentive mechanisms, each with its own miners, validators, and weight-setting, sharing one alpha token and one emission stream split by a governance parameter.

The decomposition follows the real pipeline. Building a ternary model from an FP16 parent involves a gradient-free stage — replacing linear layers with quantized equivalents, inserting the normalization that prevents variance drift, computing group scales, and snapping weights onto the grid — and then a gradient stage, in which the constrained model is trained on general text and distilled against its frozen full-precision parent until the damage is recovered. Mechanism 2 owns the first stage. Mechanism 1 owns the second. Run in sequence, they are the whole pipeline.

Each stage has a different character, which is why they deserve separate markets. Conversion is deterministic and *general*: a good conversion container works on models it has never seen, and its output can be reproduced exactly by anyone holding the container and the seed. Training is stochastic and *specific*: it produces one excellent artifact for one base model, and its quality is bounded by how much real work the miner puts in. Conversion is the higher-risk research frontier with the larger payoff if solved; training is the reliable path to the best artifact available today. Neither is chosen for being cheap to run — cost is not a property this subnet competes on, on either side of the mechanism.

4.1 Mechanism 2 — Conversion

Miners submit code, not weights. A submission is an OCI container image whose digest is committed on-chain. The container is invoked with a full-precision base model mounted read-only, a fixed calibration corpus supplied by the subnet, a target bit budget and output format, and a fixed random seed. It writes a quantized artifact to an output path. It has no network access.

The prohibition on training is enforced by budget rather than by rule. The container receives a hard wall-clock and accelerator allowance — on the order of two GPU-hours for an 8B parent — inside which quantization-aware training on billions of tokens is simply not possible. This avoids an unenforceable line between

“calibration” and “training” while leaving miners free to use any gradient-free or layer-local optimization they can afford: error compensation, per-layer threshold search, mixed allocation, reconstruction against the parent’s activations.

Validators execute the container themselves on a fixed hardware class and evaluate what comes out. Nothing a miner asserts about its artifact is taken on trust: the weights being scored are the weights the validator generated. This is the property that makes a code competition sound, and it is available here precisely because the resource budget that rules out training also puts execution within a validator’s reach.

Generality is the scored property. Each round, containers are run against announced base models *and* against surprise base models withheld until execution. A container that encodes a solution for one specific checkpoint scores nothing on the surprise set. This is the test that separates a conversion method from a lookup table, and it is the reason the network’s output is a reusable public asset rather than a pile of artifacts.

Scoring follows Affine’s approach: a submission is ranked by Pareto dominance across the cross-product of base models and evaluation axes, so a container cannot win by sacrificing agentic performance to buy a point of MMLU. Payout follows Chronos: the top ranks receive emissions on an exponential decay rather than winner-take-all, which keeps a working miner population alive between breakthroughs. Bit budget, output format, resource budget, and determinism under the fixed seed are pass/fail gates, checked before any evaluation.

4.2 Mechanism 1 — Training

Miners submit artifacts. Given a base model designated by the team, a bit budget, and a format, a miner produces the best low-bit model it can by any means: continual pretraining under the quantization constraint, distillation from the parent or from a larger teacher, whatever combination of data and schedule it can afford. The work must be real training. A miner that locates an existing low-bit checkpoint and submits it, or that takes the reigning artifact and perturbs it, has contributed nothing, and the mechanism is built to make that path lose rather than merely to detect it after the fact.

The incentive structure follows Teutonic (SN3). A reigning artifact, the KING, is public. Miners clone it, train improvements by whatever method and on whatever infrastructure they have, and submit. A submission that beats the KING by a defined margin on held-out evaluation takes the crown. Competition is continuous rather than scheduled, and a challenger’s cost of entry is the cost of improving the current best rather than the cost of starting over.

Three requirements keep it honest about what counts as work. A submission must be accompanied by a training record — the starting checkpoint it claims to descend from, the token budget consumed, and the loss trajectory — and must be derivable from that record. The king carries the heaviest burden: to hold the crown its recipe must reproduce the submitted artifact within tolerance when validators re-run it, which no miner holding a downloaded checkpoint can satisfy. And a submission whose weights sit within a fingerprint threshold of the current king, or of any published low-bit model, is rejected before evaluation rather than scored, so trivial perturbation of someone else’s artifact is not a losing strategy but an invalid one.

Two further modifications harden the competition itself. Following Affine, submissions remain private until they win, so miners are not forced to publish weights that competitors would immediately fine-tune, and scoring uses Pareto dominance across axes so the crown cannot be taken by overfitting one benchmark. Following Chronos, a share of emissions is distributed to the top ranks on an exponential decay alongside the king’s share, and evaluation uses prompts generated at evaluation time, with anti-copy and leak detection stages preceding the ranking.

4.3 Composition

The team designates base models for both mechanisms. It may designate the winning artifact of Mechanism 2 as the base model for Mechanism 1. Architecturally this requires nothing new — it is an ordinary base-model assignment — but it is the configuration in which the subnet is more than the sum of its parts.

When it is used, the network runs the full state-of-the-art pipeline end to end, with each stage independently and competitively optimized. A better initialization from Mechanism 2 means Mechanism 1 spends less of its training budget recovering and more of it improving, so the same miner compute buys a better final model. In the other direction, Mechanism 1’s results reveal which properties of an initialization actually survive training — which is information Mechanism 2 cannot generate on its own, because a raw converted model’s benchmark scores are a poor predictor of how well it trains. Each mechanism makes the other’s search more efficient, and both ratchets are public.

5. Economics

Emissions divide between the two mechanisms by a parameter the team controls and can rebalance as the network matures. A reasonable starting point weights training more heavily, since it produces the flagship artifacts, while leaving conversion a share large enough that a serious research effort against an unsolved problem is worth a miner’s time. The split is a lever on where the network directs its search, not on how much the network costs to run.

Within Bittensor’s standard split, the subnet owner receives 18% of alpha emissions, with miners and validators taking 41% each. Revenue from external customers buys and locks alpha, tying the token directly to demand for the network’s output rather than to emissions alone. Where miners are charged fees — for container execution beyond a free allowance, for instance — those fees are burned or routed to the subnet pool, never paid to validators directly, which would create a bribery channel between the scored and the scorer.

6. What the subnet sells

The recipes and artifacts are public, and that is not an obstacle to revenue. PrismML’s models are Apache 2.0 and its method is described in public literature, and Apple is evaluating it anyway — because what a buyer purchases is execution, freshness, and proof, not knowledge.

The subnet’s open catalog of best-available low-bit models is free, and functions as the network’s proof of work and its distribution. Revenue comes from running the current best method on models that are not public, which is a paid job every time and unaffected by the method being open; from the freshness of the catalog itself, since only the network knows which method is currently winning and can convert a new open release within days of publication; from independent certification, because a reproducible score from a network that does not own the model is worth more to an OEM than a vendor’s own benchmark table; and from an exclusivity window in which a winning recipe is licensable through the subnet before it enters the public commons. Serving the winning artifacts is a later addition best done through an existing inference subnet rather than built.

7. Roadmap

The first phase establishes credibility on a single, legible target: beating Bonsai-8B at its own bit budget, with both mechanisms pointed at it — Mechanism 2 attempting it by conversion alone, which no one has yet achieved at this bit depth, and Mechanism 1 attempting it by training. Independently reproducing PrismML’s published numbers under the subnet’s own harness comes before either, as the baseline and the first public artifact.

The second phase widens to continuous coverage: every significant open release converted and trained within days, across three bit tiers, with the catalog and leaderboard public and the surprise-model test keeping conversion honest as the model population grows.

The third phase opens the mechanisms to external work — private models submitted by paying customers, routed through the same machinery with a job fee attached and an emission boost, plus certification of models the network did not produce.

The fourth phase extends beyond text, since the same decomposition applies unchanged to vision, speech, and image models, where the compression pressure is if anything higher and the current low-bit quality lower.

8. Long-term vision

Model capability keeps moving, and so does the hardware it must run on. Every new frontier-class open model creates fresh demand for a compressed version on every device class that exists, and every new device class creates fresh demand for compressed versions of every model that matters. That demand does not resolve into a one-time engineering project; it is a permanent, recurring obligation that someone has to meet continuously.

Today it is met by a small number of centralized labs converting a handful of models each and negotiating bilaterally with device makers. This produces narrow coverage, opaque quality claims, and access gated by commercial relationships. The alternative is an open market where the best conversion method and the best training method are continuously contested, publicly measured, and freely available — where the question “what is the best 1.58-bit version of this model, and how good is it really” has a verifiable answer maintained by a network rather than asserted by a vendor.

That is what this subnet is for. The two mechanisms are the means: one holds a market for the method, the other a market for the artifact, and together they compound. The long-term position is to be the compression layer for open AI — the default path from any full-precision model to the best version of it that fits on the device in front of you, with a proof attached.